

Computing

Computing is any goal-oriented **ACTIVITY** requiring or creating a mathematical sequence of steps known as an **ALGORITHM**. Computing includes designing, developing and building hardware and software systems; processing, structuring, and managing various kinds of information; doing scientific research on and with computers; making computer systems behave intelligently; and creating and using communications and entertainment media. The field of computing includes computer engineering, software engineering, computer science, information systems, and information technology. Most individuals use some form of computing every day whether they realize it or not. Using ATM card, sending an email, or using a cell phone can all be considered forms of computing.

History of computing

The history of computing is longer than the history of computing hardware and modern computing technology. It covers the developments from **early simple devices to aid calculation to modern day computers**.

Prehistory of computing

Computing used during early period of civilization are categorized under prehistory computing. The main problem in early days used to be following three:

1. The development of symbolic representations of information that are amenable to manipulation.
2. The formulation of algorithms by which such representations may be processed to solve computational problems.
3. The construction of mechanical aids that allow such algorithms to be implemented in an efficient and a reliable manner

It ought to be clear that methods for representing information must have been developed first. Following table discusses the time and development of events:

Date	Event
80000 BC	Two notched rib bones may have been used for counting but they could have been merely decorative.
18000 BC	The Ishango bone, may indicate that – even so early – material objects were used for simple arithmetical operations, and it may provide evidence of some knowledge of prime numbers (although this is disputed).

2400 BC	The abacus – the first known calculator, was probably invented by the Babylonians as an aid to simple arithmetic around this time period. This laid the foundations for positional notation and later computing developments.
1115 BC	The south-pointing chariot was invented in ancient China. It was the first known geared mechanism to use a differential gear. The chariot was a two-wheeled vehicle, upon which is a pointing figure connected to the wheels by means of differential gearing. Through careful selection of wheel size, track and gear ratios, the figure atop the chariot always pointed in the same direction.
500 BC	First known use of zero by mathematicians in ancient India around this date.
500 BC	Indian grammarian Pāṇini formulated the grammar of Sanskrit (in 3959 rules) known as the Ashtadhyayi which was highly systematised and technical. Pāṇini used meta rules, transformations, and recursions with such sophistication that his grammar had the computing power equivalent to a Turing machine. Pāṇini's work was the forerunner to modern formal language theory, and a precursor to its use in modern computing. The Panini-Backus form used to describe most modern programming languages is also significantly similar to Pāṇini's grammar rules.
300 BC	Indian mathematician/scholar/musician Pingala first described the binary number system which is now used in the design of essentially all modern computing equipment. He also conceived the notion of a binary code similar to the Morse code.
200 BC	The Chinese invented the suanpan (Chinese abacus) which was widely used until the invention of the modern calculator, and continues to be used in some cultures today.
125 BC	The Antikythera mechanism: A clockwork, analog computer believed to have been designed and built in the Corinthian colony of Syracuse. The mechanism contained a differential gear and was capable of tracking the relative positions of all then-known heavenly bodies.

100 BC	Chinese mathematicians first used negative numbers .
60	Heron of Alexandria made numerous inventions, including "Sequence Control" in which the operator of a machine set a machine running, which then follows a series of instructions in a deterministic fashion. This was, essentially, the first program. He also made numerous innovations in the field of automata, which are important steps in the development of robotics.
200	Indian Jaina mathematicians invented logarithms .
600	Indian mathematician Brahma Gupta was the first to describe the modern place-value numeral system (Hindu numeral system).
724	Chinese inventor Liang Lingzan built the world's first fully mechanical clock ; water clocks, some of them extremely accurate, had been known for centuries previous to this. This was an important technological leap forward; the earliest true computers, made a thousand years later, used technology based on that of clocks.
820	Persian mathematician, Muḥammad ibn Mūsā al-Khwārizmī, described the rudiments of modern algebra whose name is derived from his book <i>Al-Kitāb al-muḥtaṣar fī ḥisāb al-ğabr wa-l-muqābala</i> . The word <i>algorithm</i> is derived from al-Khwarizmi's Latinized name <i>Algoritmi</i> .
850	Arab mathematician, Al-Kindi (Alkindus), was a pioneer of cryptography . He gave the first known recorded explanation of cryptanalysis in <i>A Manuscript on Deciphering Cryptographic Messages</i> . In particular, he is credited with developing the frequency analysis method whereby variations in the frequency of the occurrence of letters could be analyzed and exploited to break encryption ciphers (i.e. cryptanalysis by frequency analysis). The text also covers methods of cryptanalysis, encipherments, cryptanalysis of certain encipherments, and statistical analysis of letters and letter combinations in Arabic.

850	The Banū Mūsā brothers, in their <i>Book of Ingenious Devices</i> , invented "the earliest known mechanical musical instrument", in this case a hydro powered organ which played interchangeable cylinders automatically. This "cylinder with raised pins on the surface remained the basic device to produce and reproduce music mechanically until the second half of the nineteenth century. They also invented an automatic flute player which appears to have been the first programmable machine.
996	Persian astronomer, Abū Rayhān al-Bīrūnī, invented the first geared mechanical astrolabe, featuring eight gear-wheels. This can be considered as an ancestor of the mechanical clocks.
1000	Abū Rayhān al-Bīrūnī invented the Planisphere, an analog computer. ^[citation needed] He also invented the first mechanical lunisolar calendar which employed a gear train and eight gear-wheels. This was an early example of a fixed-wired knowledge processing machine.
1015	Arab astronomer, Abū Ishāq Ibrāhīm al-Zarqālī (Arzachel) of al-Andalus, invented the Equatorium, a mechanical analog computer device used for finding the longitudes and positions of the Moon, Sun and planets without calculation, using a geometrical model to represent the celestial body's mean and anomalistic position. ^[11]
1020	The mechanical geared astrolabe earlier developed by Abū Rayhān al-Bīrūnī perfected by Ibn Samh. This can be considered an ancestor of the mechanical clock.
1100	Arab astronomer, Jabir ibn Aflah (Geber), invented the Torquetum, an observational instrument and mechanical analog computer device used to transform between spherical coordinate systems. It was designed to take and convert measurements made in three sets of coordinates: horizon, equatorial, and ecliptic.
1206	Arab engineer, Al-Jazari, invented numerous automata and made numerous other technological innovations. One of these is a design for a programmable humanoid-shaped mannequin: this seems to have been the first serious, scientific (as opposed to

	magical) plan for a robot.^[13] He also invented the "castle clock", an astronomical clock which is considered to be the earliest programmable analog computer. It displayed the zodiac, the solar and lunar orbits, a crescent moon-shaped pointer travelling across a gateway causing automatic doors to open every hour, and five robotic musicians who play music when struck by levers operated by a camshaft attached to a water wheel. The length of day and night could be re-programmed every day in order to account for the changing lengths of day and night throughout the year.
1235	Persian astronomer Abi Bakr of Isfahan invented a brass astrolabe with a geared calendar movement based on the design of Abū Rayhān al-Bīrūnī's mechanical calendar analog computer.^[17] Abi Bakr's geared astrolabe uses a set of gear-wheels and is the oldest surviving complete mechanical geared machine in existence.
1300	Ramon Llull invented the Lullian Circle: a notional machine for calculating answers to philosophical questions (in this case, to do with Christianity) via logical combinatorics. This idea was taken up by Leibniz centuries later, and is thus one of the founding elements in computing and information science
1400	Kerala school of astronomy and mathematics in South India invented the floating point number system.
1400	Jamshīd al-Kāshī invented the <i>Plate of Conjunctions</i>, an analog computer instrument used to determine the time of day at which planetary conjunctions will occur, and for performing linear interpolation. He also invented a mechanical "planetary computer" which he called the <i>Plate of Zones</i>, which could graphically solve a number of planetary problems, including the prediction of the true positions in longitude of the Sun and Moon, and the planets; the latitudes of the Sun, Moon, and planets; and the ecliptic of the Sun. The instrument also incorporated an alidade and ruler.
1400	Ahmad al-Qalqashandi gives a list of ciphers in his <i>Subh al-a'sha</i> which include both substitution and transposition, and for the first time, a cipher with multiple substitutions for each plaintext letter. He also gives an exposition on and worked

	example of cryptanalysis, including the use of tables of letter frequencies and sets of letters which cannot occur together in one word.
1492	Leonardo da Vinci produced drawings of a device consisting of interlocking cog wheels which can be interpreted as a mechanical calculator capable of addition and subtraction. A working model inspired by this plan was built in 1968 but it remains controversial whether Leonardo really had a calculator in mind. Da Vinci also made plans for a mechanical man: an early design for a robot.
1588	Joost Buerghi discovered natural logarithms .
1614	Scotsman John Napier reinvented a form of logarithms and an ingenious system of movable rods (referred to as Napier's Rods or Napier's bones). These rods were based on the lattice or gelosia multiplication algorithm and allowed the operator to multiply, divide and calculate square and cube roots by moving the rods around and placing them in specially constructed boards.
1622	William Oughtred developed slide rules based on natural logarithms as developed by John Napier.
1623	German polymath Wilhelm Schickard drew a device that he called a <i>Calculating Clock</i> on two letters that he sent to Johannes Kepler; one in 1623 and the other in 1624. A fire later destroyed the machine as it was being built in 1624 and he decided to abandon his project. This machine became known to the world only in 1957 when the two letters were discovered. Some replicas were built in 1961. This machine had no impact on the development of mechanical calculators.

History of Computer hardware

The term "computer", in use from the early 17th meant "one who computes": a person performing mathematical calculations, before electronic computers became commercially available. "The human computer is supposed to be following fixed rules; he has no authority to deviate from them in any detail."

Once proper physical computing device was developed, then this terminology no longer indicated person. These physical devices, computer, has gone under tremendous changes since its invention. Thus, it is termed as evolution of computer some time. Below is the table that describes the major changes in computer hardware in its history:

1822	Charles Babbage designed his first mechanical computer, the first prototype of the decimal difference engine for tabulating polynomials.
1832	Semen Korsakov proposed the usage of punched cards for information storage and search. He designed several machines to demonstrate his ideas, including the so-called <i>linear homeoscope</i> .
1832	Babbage and Joseph Clement produced a prototype segment of his difference engine, which operated on 6-digit numbers and second-order differences (i.e., it could tabulate quadratic polynomials). The complete engine, which would have been room-sized, was planned to operate both on sixth-order differences with numbers of about 20 digits, and on third-order differences with numbers of 30 digits. Each addition would have been done in two phases, the second one taking care of any carries generated in the first. The output digits were to be punched into a soft metal plate, from which a printing plate might have been made. But there were various difficulties, and no more than this prototype piece was ever finished.
1834	Babbage conceived, and began to design, his decimal 'Analytical Engine'. A program for it was to be stored on read-only memory, in the form of punched cards. Babbage continued to work on the design for years, though after about 1840 design changes seem to have been minor. The machine would have operated on 40-digit numbers; the 'mill' (CPU) would have had 2 main accumulators and some auxiliary ones for specific purposes, while the 'store' (memory) would have held a thousand 50-digit numbers. There would have been several punched card readers, for both programs and data; the cards were to be chained and the motion of each chain reversible. The machine would have performed conditional jumps. There would also have been a form of microcoding: the meaning of instructions were to depend on the positioning of metal studs in a slotted barrel, called the "control barrel". The machine envisioned would have been capable of an addition in 3 seconds and a multiplication or division in 2–4 minutes. It was to be powered by a steam engine. In the end, no more than a few parts were actually built.

1843	Per Georg Scheutz and his son Edvard produced a third-order difference engine with printer; the Swedish government agrees to fund their next development.
1847	Babbage designed an improved, simpler difference engine (the Difference Engine No.2), a project which took 2 years. The machine would have operated on 7th-order differences and 31-digit numbers, but nobody was found to pay to have it built. In 1989-1991 a team at London's Science Museum did build one from the surviving plans. They built components using modern methods, but with tolerances no better than Clement could have provided... and, after a bit of tinkering and detail-debugging, they found that the machine works properly. In 2000, the printer was also completed.
1848	British Mathematician George Boole developed binary algebra (Boolean algebra) which has been widely used in binary computer design and operation, beginning about a century later. See 1939.
1851	After 30 years of development, Thomas de Colmar launched the mechanical calculator industry by starting the manufacturing of a much simplified Arithmometer (invented in 1820). Aside from its clones, which started thirty years later, ^[36] it was the only calculating machine available anywhere in the world for forty years (Dorr E. Felt only sold one hundred comptometers and a few comptographs from 1887 to 1890 ^[37]). Its simplicity made it the most reliable calculator to date. It was a big machine (a 20 digit arithmometer was long enough to occupy most of a desktop). Even though the arithmometer was only manufactured until 1915, twenty European companies manufactured improved clones of its design until the beginning of WWII ; they were Burkhardt, Layton, Saxonia, Gräber, Peerless, Mercedes-Euklid, XxX, Archimedes, etc...
1853	To Babbage's delight, the Scheutzes completed the first full-scale difference engine, which they called a Tabulating Machine. It operated on 15-digit numbers and 4th-order differences, and produced printed output just as Babbage's would have. A second machine was later built to the same design by the firm of Bryan Donkin of London.

1906	Vacuum tube (or thermionic valve) invented by Lee De Forest.
1919	William Henry Eccles and F. W. Jordan published the first flip-flop circuit design.
1924	Walther Bothe built an AND logic gate - the <i>coincidence circuit</i> , for use in physics experiments, for which he received the Nobel Prize in Physics 1954. Digital circuitries of all kinds make heavy use of this technique.
1928	IBM standardizes on punched cards with 80 columns of data and rectangular holes. Widely known as IBM Cards, they dominate the data processing industry for almost half a century.
1930	Vannevar Bush built a partly electronic difference engine capable of solving differential equations.
1931	IBM introduced the IBM 601 Multiplying Punch, an electromechanical machine that could read two numbers, up to 8 digits long, from a card and punch their product onto the same card. ^[45]
1938	Konrad Zuse of Berlin, completed the 'Z1', the first mechanical binary programmable computer. It was based on Boolean Algebra and had some of the basic ingredients of modern machines, using the binary system and floating-point arithmetic. Zuse's 1936 patent application (Z23139/GMD Nr. 005/021) also suggested a 'von Neumann' architecture (re-invented about 1945) with program and data modifiable in storage. Originally the machine was called the 'V1' but retroactively renamed after the war, to avoid confusion with the V-1 flying bomb. It worked with floating point numbers (7-bit exponent, 16-bit mantissa, and sign bit). The memory used sliding metal parts to store 16 such numbers, and worked well; but the arithmetic unit was less successful, occasionally suffering from certain mechanical engineering problems. The program was read from holes punched in discarded 35 mm movie film. Data values could have been entered from a numeric keyboard, and outputs were displayed on electric lamps. The

	machine was not a general purpose computer (i.e., Turing complete) because it lacked loop capabilities.
1939	Konrad Zuse completed the 'Z2' (originally 'V2'), which combined the Z1's existing mechanical memory unit with a new arithmetic unit using relay logic. Like the Z1, the Z2 lacked loop capabilities. The project was interrupted for a year when Zuse was drafted, but continued after he was released.
1941 May 11	Now working with limited backing from the DVL (German Aeronautical Research Institute), Konrad Zuse completed the ' Z3 ' (originally 'V3'): the first operational programmable computer. One major improvement over Charles Babbage's non-functional device is the use of Leibniz's binary system (Babbage and others unsuccessfully tried to build decimal programmable computers). Zuse's machine also featured floating point numbers with a 7-bit exponent, 14-bit mantissa (with a '1' bit automatically prefixed unless the number is 0), and a sign bit. The memory held 64 of these words and therefore required over 1400 relays; there were 1200 more in the arithmetic and control units. It also featured parallel adders. The program, input, and output were implemented as described above for the Z1. Although conditional jumps were not available, it has been shown that Zuse's Z3 is, in principle, capable of functioning as a universal computer. ^{[48][49]} The machine could do 3-4 additions per second, and took 3–5 seconds for a multiplication. The Z3 was destroyed in 1943 during an Allied bombardment of Berlin, and had no impact on computer technology in America and England.
1942	Konrad Zuse developed the S1, the world's first process computer, used by Henschel to measure the surface of wings.
1944 Aug 7	The IBM Automatic Sequence Controlled Calculator was turned over to Harvard University, which called it the Harvard Mark I. It was designed by Howard Aiken and his team, financed and built by IBM—it became the second program controlled machine (after Konrad Zuse's). The whole machine was 51 feet (16 m) long, weighed 5 (short) tons (4.5 tonnes), and incorporated 750,000 parts. It used 3304 electromechanical relays as on-off switches, had 72 accumulators (each with its own arithmetic unit), as well as a mechanical register with a capacity of 23 digits plus sign. The arithmetic was fixed-point and decimal, with a control panel setting determining the number of decimal places.

	Input-output facilities include card readers, a card punch, paper tape readers, and typewriters. There were 60 sets of rotary switches, each of which could be used as a constant register —sort of mechanical read-only memory. The program was read from one paper tape; data could be read from the other tapes, or the card readers, or from the constant registers. Conditional jumps were not available. However, in later years, the machine was modified to support multiple paper tape readers for the program, with the transfer from one to another being conditional, rather like a conditional subroutine call. Another addition allowed the provision of plug-board wired subroutines callable from the tape. Used to create ballistics tables for the US Navy.
1945	John von Neumann drafted a report describing the future computer eventually built as the EDVAC (Electronic Discrete Variable Automatic Computer). <i>First Draft of a Report on the EDVAC</i> includes the first published description of the design of a stored-program computer, giving rise to the term von Neumann architecture. It directly or indirectly influenced nearly all subsequent projects, especially EDSAC. The design team included John W. Mauchly and J. Presper Eckert.
1946 Feb 14	ENIAC (Electronic Numerical Integrator and Computer): One of the first totally electronic, valve driven, digital, program-controlled computers was unveiled although it was shut down on 9 November 1946 for a refurbishment and a memory upgrade, and was transferred to Aberdeen Proving Ground, Maryland in 1947. Development had started in 1943 at the Ballistic Research Laboratory, USA, by John W. Mauchly and J. Presper Eckert. It weighed 30 tonnes and contained 18,000 electronic valves, consuming around 160 kW of electrical power. It could do 50,000 basic calculations a second. It was used for calculating ballistic trajectories and testing theories behind the hydrogen bomb.
1946 Feb 19	ACE (Automatic Computing Engine): Alan Turing presented a detailed paper to the National Physical Laboratory (NPL) Executive Committee, giving the first reasonably complete design of a stored-program computer. However, because of the strict and long-lasting secrecy around his wartime work at Bletchley Park, he was prohibited (having signed the Official Secrets Act) from explaining that he knew that his ideas could be implemented in an electronic device.
1946	The trackball was invented as part of a radar plotting system named Comprehensive Display System (CDS) by Ralph Benjamin when working for the British Royal

	Navy Scientific Service.^{[54][55]} Benjamin's project used analog computers to calculate the future position of target aircraft based on several initial input points provided by a user with a joystick. Benjamin felt that a more elegant input device was needed and invented a <i>ball tracker</i>^{[54][55]} system called the <i>roller ball</i>^[54] for this purpose in 1946.^{[54][55]} The device was patented in 1947,^[54] but only a prototype was ever built^[55] and the device was kept as a secret outside military.^[55]
1947 Dec 16	Invention of the transistor at Bell Laboratories, USA, by William B. Shockley, John Bardeen and Walter Brattain.
1947	Howard Aiken completed the Harvard Mark II.
1948 Jan 27	IBM finished the SSEC (Selective Sequence Electronic Calculator). It was the first computer to modify a stored program. "About 1300 vacuum tubes were used to construct the arithmetic unit and eight very high-speed registers, while 23000 relays were used in the control structure and 150 registers of slower memory."
1948 Jul 21	SSEM, Small-Scale Experimental Machine or 'Baby' was built at the University of Manchester. It ran its first program on this date. It was the first computer to store both its programs and data in RAM, as modern computers do. By 1949 the 'Baby' had grown, and acquired a magnetic drum for more permanent storage, and it became the Manchester Mark 1.
1948	ANACOM from Westinghouse was an AC-energized electrical analog computer system used up until the early 1990s for problems in mechanical and structural design, fluidics, and various transient problems.
1949 Mar	John Presper Eckert and John William Mauchly construct the BINAC for Northrop.

1949 May 6	This is considered the birthday of modern computing. Maurice Wilkes and a team at Cambridge University executed the first stored program on the EDSAC computer, which used paper tape input-output. Based on ideas from John von Neumann about stored program computers, the EDSAC was the first complete, fully functional von Neumann architecture computer.
1949 Oct	The Manchester Mark 1 final specification is completed; this machine was notably in being the first computer to use the equivalent of base/index registers, a feature not entering common computer architecture until the second generation around 1955.
1949	CSIR Mk I (later known as CSIRAC), Australia's first computer, ran its first test program. It was a vacuum tube based electronic general purpose computer. Its main memory stored data as a series of acoustic pulses in 5 ft (1.5 m) long tubes filled with mercury.
1949	MONIAC (Monetary National Income Analogue Computer) also known as the Phillips Hydraulic Computer, was created in 1949 to model the national economic processes of the United Kingdom. The MONIAC consisted of a series of transparent plastic tanks and pipes. It is thought that twelve to fourteen machines were built.
1949	“ Computers in the future may weigh no more than 1.5 tons. ” — <i>Popular Mechanics</i>, forecasting the relentless march of science

Generations of computers

Time	Technology	Size and Speed	Language	Example	Other Features
1940's-1950's	Vacuum Tubes	Huge, taking up entire rooms, Slow speed	Machine Language	UNIVAC and ENIAC	Used a great deal of electricity. Generated a lot of heat. Input was based on punched cards and paper tape, and output was displayed on printouts.

1950's-1960's	Transistors	Lesser size and increased speed	Assembly language and languages like COBOL and FORTRAN	UNIVAC 1108, IBM 1401, CDC 1604	More efficient and reliable. Though the transistors still generated a great deal of heat that subjected the computer to damage, it was a vast improvement over the vacuum tube. Second-generation computers still relied on punched cards for input and printouts for output.
late 1960's-1970's	Integrated Circuit	Size Lesser and speed further increased	OS was developed	IBM-360 series, Honeywell Model 316, Honeywell – 6000 series, CDC – 1700.	Instead of punched cards and printouts, users interacted with third generation computers through keyboards and monitors and interfaced with an operating system, which allowed the device to run many different applications at one time with a central program that monitored the memory. Computers for the first time became accessible to a mass audience because they were smaller and cheaper than their predecessors.
1970's-today	Microprocessor	Reduced size and tremendous speed	High Level Languages like PASCAL, COBOL, C, C++, JAVA	Intel 4004, Apple Macintosh	Microprocessors also moved out of the realm of desktop computers and into many areas of life as more and more everyday products began to use microprocessors. As these small computers became more powerful, they could be linked together to form networks, which eventually led to the development of the Internet. Fourth generation computers also saw the development of GUIs, the mouse and handheld devices.

1990's -today	Microprocessor	Reduced size and tremendous speed	Based on Artificial intelligence	Parallel Inference Machine	Fifth generation computing devices, based on artificial intelligence, are still in development, though there are some applications, such as voice recognition, that are being used today. The goal of fifth-generation computing is to develop devices that respond to natural language input and are capable of learning and self-organization.
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History of programming language

Ever since the invention of Charles Babbage's difference engine in 1822, computers have required a means of instructing them to perform a specific task. This means is known as a programming language. Computer languages were first composed of a series of steps to wire a particular program; these morphed into a series of steps keyed into the computer and then executed; later these languages acquired advanced features such as logical branching and object orientation.

In the beginning, Charles Babbage's difference engine could only be made to execute tasks by changing the gears which executed the calculations. Thus, the earliest form of a computer language was physical motion. Eventually, physical motion was replaced by electrical signals when the US Government built the ENIAC in 1942. It followed many of the same principles of Babbage's engine and hence, could only be "programmed" by presetting switches and rewiring the entire system for each new "program" or calculation. This process proved to be very tedious.

In 1945, John Von Neumann was working at the Institute for Advanced Study. He developed two important concepts that directly affected the path of computer programming languages. The first was known as "shared-program technique". This technique stated that the actual computer hardware should be simple and not need to be hand-wired for each program. Instead, complex instructions should be used to control the simple hardware, allowing it to be reprogrammed much faster.

The second concept was also extremely important to the development of programming languages. Von Neumann called it "conditional control transfer". This idea gave rise to the notion of subroutines, or small blocks of code that could be jumped to in any order, instead of a single set of chronologically ordered steps for the computer to take. The second part of the idea stated that computer code should be able to branch based on logical statements such as IF (expression) THEN, and looped such as with a FOR statement. "Conditional control transfer" gave rise to the idea of "libraries," which are blocks of code that can be reused over and over. Around this time, Konrad Zuse, a German, was inventing his own computing systems independently and developed many of

the same concepts, both in his machines and in the Plankalkul programming language. Alas, his work did not become widely known until much later.

In 1949, a few years after Von Neumann's work, the language Short Code appeared. It was the first computer language for electronic devices and it required the programmer to change its statements into 0's and 1's by hand. Still, it was the first step towards the complex languages of today. In 1951, Grace Hopper wrote the first compiler, A-0. A compiler is a program that turns the language's statements into 0's and 1's for the computer to understand. This led to faster programming, as the programmer no longer had to do the work by hand.

In 1957, the first of the major languages appeared in the form of FORTRAN. Its name stands for FORMula TRANslating system. The language was designed at IBM for scientific computing. The components were very simple, and provided the programmer with low-level access to the computers innards. Today, this language would be considered restrictive as it only included IF, DO, and GOTO statements, but at the time, these commands were a big step forward. The basic types of data in use today got their start in FORTRAN, these included logical variables (TRUE or FALSE), and integer, real, and double-precision numbers.

Though FORTRAN was good at handling numbers, it was not so good at handling input and output, which mattered most to business computing. Business computing started to take off in 1959, and because of this, COBOL was developed. It was designed from the ground up as the language for businessmen. Its only data types were numbers and strings of text. It also allowed for these to be grouped into arrays and records, so that data could be tracked and organized better. It is interesting to note that a COBOL program is built in a way similar to an essay, with four or five major sections that build into an elegant whole. COBOL statements also have a very English-like grammar, making it quite easy to learn. All of these features were designed to make it easier for the average business to learn and adopt it.

In 1958, John McCarthy of MIT created the LISP (or LISP) language. It was designed for Artificial Intelligence (AI) research. Because it was designed for a specialized field, the original release of LISP had a unique syntax: essentially none. Programmers wrote code in parse trees, which are usually a compiler-generated intermediary between higher syntax (such as in C or Java) and lower-level code. Another obvious difference between this language (in original form) and other languages is that the basic and only type of data is the list; in the mid-1960's, LISP acquired other data types. A LISP list is denoted by a sequence of items enclosed by parentheses. LISP programs themselves are written as a set of lists, so that LISP has the unique ability to modify itself, and hence grow on its own. The LISP syntax was known as "Cambridge Polish," as it was very different from standard Boolean logic:

The Algol language was created by a committee for scientific use in 1958. Its major contribution is being the root of the tree that has led to such languages as Pascal, C, C++, and Java. It was also the first language with a formal grammar, known as Backus-Naar Form or BNF. Though Algol

implemented some novel concepts, such as recursive calling of functions, the next version of the language, Algol 68, became bloated and difficult to use. This led to the adoption of smaller and more compact languages, such as Pascal.

Pascal was begun in 1968 by Niklaus Wirth. Its development was mainly out of necessity for a good teaching tool. In the beginning, the language designers had no hopes for it to enjoy widespread adoption. Instead, they concentrated on developing good tools for teaching such as a debugger and editing system and support for common early microprocessor machines which were in use in teaching institutions.

Pascal was designed in a very orderly approach, it combined many of the best features of the languages in use at the time, COBOL, FORTRAN, and ALGOL. While doing so, many of the irregularities and oddball statements of these languages were cleaned up, which helped it gain users. The combination of features, input/output and solid mathematical features, made it a highly successful language. Pascal also improved the “pointer” data type, a very powerful feature of any language that implements it. It also added a CASE statement that allowed instructions to branch like a tree in such a manner.

Pascal also helped the development of dynamic variables, which could be created while a program was being run, through the NEW and DISPOSE commands. However, Pascal did not implement dynamic arrays, or groups of variables, which proved to be needed and led to its downfall. Wirth later created a successor to Pascal, Modula-2, but by the time it appeared, C was gaining popularity and users at a rapid pace.

C was developed in 1972 by Dennis Ritchie while working at Bell Labs in New Jersey. The transition in usage from the first major languages to the major languages of today occurred with the transition between Pascal and C. Its direct ancestors are B and BCPL, but its similarities to Pascal are quite obvious. All of the features of Pascal, including the new ones such as the CASE statement are available in C. C uses pointers extensively and was built to be fast and powerful at the expense of being hard to read. But because it fixed most of the mistakes Pascal had, it won over former-Pascal users quite rapidly.

Ritchie developed C for the new Unix system being created at the same time. Because of this, C and Unix go hand in hand. Unix gives C such advanced features as dynamic variables, multitasking, interrupt handling, forking, and strong, low-level, input-output. Because of this, C is very commonly used to program operating systems such as Unix, Windows, the MacOS, and Linux.

In the late 1970's and early 1980's, a new programming method was being developed. It was known as Object Oriented Programming, or OOP. Objects are pieces of data that can be packaged and manipulated by the programmer. Bjarne Stroustrup liked this method and developed extensions to C known as “C with Classes.” This set of extensions developed into the full-featured language C++, which was released in 1983.

C++ was designed to organize the raw power of C using OOP, but maintain the speed of C and be able to run on many different types of computers. C++ is most often used in simulations, such as games. C++ provides an elegant way to track and manipulate hundreds of instances of people in elevators, or armies filled with different types of soldiers. It is the language of choice in today's AP Computer Science courses.

In the early 1990's, interactive TV was the technology of the future. Sun Microsystems decided that interactive TV needed a special, portable (can run on many types of machines), language. This language eventually became Java. In 1994, the Java project team changed their focus to the web, which was becoming "the cool thing" after interactive TV failed. The next year, Netscape licensed Java for use in their internet browser, Navigator. At this point, Java became the language of the future and several companies announced applications which would be written in Java, none of which came into use.

Though Java has very lofty goals and is a text-book example of a good language, it may be the "language that wasn't." It has serious optimization problems, meaning that programs written in it run very slowly. And Sun has hurt Java's acceptance by engaging in political battles over it with Microsoft. But Java may wind up as the instructional language of tomorrow as it is truly object-oriented and implements advanced techniques such as true portability of code and garbage collection.

Visual Basic is often taught as a first programming language today as it is based on the BASIC language developed in 1964 by John Kemeny and Thomas Kurtz. BASIC is a very limited language and was designed for non-computer science people. Statements are chiefly run sequentially, but program control can change based on IF..THEN, and GOSUB statements which execute a certain block of code and then return to the original point in the program's flow.

Microsoft has extended BASIC in its Visual Basic (VB) product. The heart of VB is the form, or blank window on which you drag and drop components such as menus, pictures, and slider bars. These items are known as "widgets." Widgets have properties (such as its color) and events (such as clicks and double-clicks) and are central to building any user interface today in any language. VB is most often used today to create quick and simple interfaces to other Microsoft products such as Excel and Access without needing a lot of code, though it is possible to create full applications with it.

Perl has often been described as the "duct tape of the Internet," because it is most often used as the engine for a web interface or in scripts that modify configuration files. It has very strong text matching functions which make it ideal for these tasks. Perl was developed by Larry Wall in 1987 because the Unix sed and awk tools (used for text manipulation) were no longer strong enough to support his needs. Depending on whom you ask, Perl stands for Practical Extraction and Reporting Language or Pathologically Eclectic Rubbish Lister.

Programming languages have been under development for years and will remain so for many years to come. They got their start with a list of steps to wire a computer to perform a task. These steps eventually found their way into software and began to acquire newer and better features. The first major languages were characterized by the simple fact that they were intended for one purpose and

one purpose only, while the languages of today are differentiated by the way they are programmed in, as they can be used for almost any purpose. And perhaps the languages of tomorrow will be more natural with the invention of quantum and biological computers.

History of operating systems

Historically operating systems have been tightly related to the computer architecture. Operating systems have evolved through a number of distinct phases or generations which corresponds roughly to the decades.

The 1940's

Computers were a new idea. Typical general purpose computer designs in this era were closely modeled on the paper design presented at the Princeton summer school in 1946. That means a single-accumulator machine with 40 bits per word, with two 20 bit instructions packed into each word. Memory addresses were typically 12 bits, allowing addressing of 4k words of 40 bits each. Half word addressing was supported in only the most minimal way by having two distinct jump instructions, jump to high halfword and jump to low halfword.

Subroutine call instructions had not yet been invented. To call a subroutine took, in the typical case, 3 instructions.

Load constant (the return jump) into accumulator.

Store accumulator (the return jump) into the subroutine's return location.

Jump to the entry point of the subroutine.

The return location of the subroutine was typically the last instruction of the subroutine, and the constant loaded, for each call, was not the return address, but rather, a fully formed jump instruction to transfer control to the return address.

With machines like this, it was hard to imagine any kind of operating system.

The 1950's

The decade of the 1950 saw a number of innovations in computer architecture. These included the invention of index registers and subroutine call instructions, both crucial to the development of modern ideas of programming. This was also the decade when the first compilers were developed, most notably, FORTRAN.

In the 1950's, magnetic tape drives became commonplace. Typical tape drives for computers stored data as a sequence of records. Each record was typically 80 characters (based on the record format of punched cards) or 120 characters (the number of characters per line on many early line printers), although the hardware did not limit the record format.

Computers of the 1950's rapidly standardized on 6 bits per character, after the IBM 701 computer introduced this standard. The 701 also set the word-size to 36 bits, and many others used this size, while 48-bit words were also used on some computers. The standard magnetic tape format that emerged in this era, also introduced by IBM, stored 7 data tracks on 1/2 inch wide tape. 7-tracks were used so that 6 bits of data (one character) could be stored in parallel along with a parity bit.

Data was typically stored at just 100 characters per inch (but higher density tapes came out, first 200 characters per inch, and then 400), and tape reels typically held 1200 feet of tape.

The tape drive could not read or write fractional records, and the drive hardware automatically computed (on output) or checked (on input) the parity of each character. The drive also computed or checked the checksum of the entire block, which was stored at the end of the block.

Typical tape drive commands were:

Read block into memory buffer.

Write block from memory buffer.

Skip forward one block.

Skip backward one block.

Write end-of-file mark (conceptually, a special kind of block).

Skip forward to next end of file.

Skip backward to next end of file.

Rewind tape.

By mid decade, computers with whole banks of tape drives became common. In such a mix, one tape drive was frequently designated the system drive, and the tape on that drive contained system programs. Main memories were still small, but it was common for a small loader to be permanently resident in main memory, able to load and run the n'th file from tape drive d (you'd jump to the loader with parameters n and d loaded in agreed upon memory locations or registers).

This system is a sufficient foundation to make something that users would begin to call a tape operating system. At the end of execution, typical programs would exit by jumping to the loader, asking it to load and run the command language interpreter from the system tape. The command language interpreter would remember (in just a few words of dedicated memory) the drive number from which it was reading a command file. Commands in the command file would set up parameters to programs and then launch them.

These systems were very fragile. Any program that accidentally damaged the loader would force a complete system restart. Any program that accidentally damaged the memory location used to remember the current input file could lead to wild and unpredictable system actions.

Nonetheless, these systems were flexible enough to support assemblers and compilers, linkers and subroutine libraries. Fortran grew in such an environment, and the developers thinking about the new languages of 1960, Cobol and Algol, had extensive experience with such systems.

Rotating magnetic memory was also important in this era. Not disk drives, but rather, drums. Some low performance computers had drums for main memory. Typically, drum main memory stored data in word parallel form, so a computer with a 40 bit word would have a 40 track drum, or perhaps 44 tracks, so it could include a parity bit and some tracks for addressing (typically, one track for counting words, and a track holding a start mark so it could tell where on each revolution the word count should be reset to zero).

File systems on drum computers had yet to be developed, but subroutine libraries typically included routines to read a block n words of data in from drum address d to main memory address m , or to write a block of data back out to the drum. Clever programmers could use these to move subroutines or data structures out of main memory when they were not needed, reading them back in only when needed. This was called overlay management, and it was very difficult to get it right.

The 1960's

During the 1960's, the major developments in computer architecture were condition codes, byte addressing, memory management units, and various forms of parallel processing. Parallel processing ranged from attaching multiple co-equal CPUs to a single memory through using dedicated small general purpose processors for input-output to special purpose coprocessors. The first graphics coprocessors emerged in this era, but the most common use of coprocessors was to speed input-output to the newly developed high performance moving-head disk drives.

The first real operating systems emerged in the 1960s. Some of these were very crude. The acronym DOS, standing for disk operating system, first emerged in this era.

A typical DOS involved just one change to the tape operating system described above. The system subroutine library included a file system, so that programs could use a disk drive as if it were multiple tape drives, where disk files each had textual names. The command language interpreter could read commands from a disk file, launch programs from disk files, and for each program launched, tell it what files to use.

In the DOS era, tapes were used for backups and for files that were too large to fit on the disk drives.

The first networking efforts emerged in this era, the and the dial-up modem came into being -- typically, dial-up access was at 110 baud for electromechanical Teletype terminals, but the first generation of modems were designed to work at up to 300 baud.

Memory Management Units

Memory management units were introduced very early in the 1960s by Feranti corporation on their Atlas computer. The Atlas system had paged virtual memory, and the Atlas operating system used it for both memory protection and to create the illusion of a large address space implemented using a small main memory and what was, at the time, a large magnetic drum.

By the end of the decade, IBM would release a computer, the IBM 360 model 67 that supported this technology but several manufacturers got there first, including Scientific Data Systems (the SDS 940), General Electric (the GE 600) and Digital Equipment Corporation (the PDP-10). All of the latter virtual memory systems were built by customers (the University of California at Berkeley, MIT and Bolt, Beranek and Newman), but were then sold commercially.

Memory management units required real operating systems. The University of California at Berkeley developed the Berkeley Timesharing System for the SDS 940. General Electric, in conjunction with Bell Labs and MIT developed Multics for the GE 600, and BB&N developed TENEX for the PDP-10. IBM's first attempt at an operating system for the 360/67, TSS 360, was a failure. They never got it working (Carnegie-Mellon University, which had a 360/67 on order, took delivery on the broken TSS/360 software and fixed it). IBM's follow up, the System 370 with the VM operating system, however, was very successful in the 1970's.

Multics

Multics was, undoubtedly, the single most important operating system developed in the 1960's. After Honeywell bought GE's computer division, it became the flagship operating system of the H 6000 series of mainframes, the successor to the GE 600. Multics introduced the following ideas:

A hierarchic file system. Each user had a home directory, with their own subdirectories hanging from that home directory.

Access control lists for files.

The idea of a computer utility, where anyone could subscribe. Today, we call these ISPs.

Multiple levels of protection, so that lower level parts of the operating system were protected from upper level parts, which were protected from user programs.

Opening a file being the same as loading the contents of that file as a memory segment.

Multics also built on some ideas that first came to market in the Berkeley Timesharing System, including

Separate virtual address spaces for each user

The ability to share memory segments between users

Multics was definitely not modern in some ways: The GE 600 and the H 6000 had 36-bit words, and the machine supported two character sets, GE's 6-bit code, packed 6 characters per word, and

7-bit ASCII. Five 7-bit characters could be packed into 35 bits (leaving the sign bit unused) or each 7-bit character could be padded out to 9 bits, packing 4 characters per word. The former made more efficient use of memory, while the latter was easier for programmers.

The 1970's

What most of the world saw in the 1970's computer market was a steep plunge in the price of computing. This actually began with the introduction of minicomputers in the 1960's, with the least expensive general purpose computer systems selling for under \$10,000 by 1970, but the trend quickly accelerated, until by the mid 1970's, a fully functional microcomputer kit could be purchased for under \$1000 in kit form (the Altair 8800).

When Bell Labs quit the Multics project, some of the programmers working on that project decided to take the best ideas they'd encountered in that project and scale them down, building a little operating system suitable for a departmental timesharing system running on a minicomputer. The result was Unix. Even the name is a pun on Multics.

It is fair to say that Unix had only one new idea -- the SUID and SGID bits on files. Everything else had been done before. What Unix did was do all of it better, integrating a number of really good ideas from multiple sources (mostly Multics and the Berkeley Timesharing System) into one system and doing it very well.

Another system from the 1970's is largely forgotten outside of corporate datacenters. That is VM-370. This is the system that IBM developed out of the ashes of the TSS 360 project on the IBM 360/67 -- specifically, the CP/CMS operating system developed in the very late 1960s by IBM. What VM did that had never been done before is virtualize everything, so that users could run any operating system they wanted as user programs under VM (originally CP). The idea of being to run, say, the horrible old DOS 360 as a user program on a computer without threatening any other user of that computer was extraordinary. The IBM 360 was the first 32-bit computer of any consequence, and IBM's current 64-bit Enterprise Systems Architecture is compatible with 360 and its successor, the 370.

Networking, introduced in the mid 1960s, became commonplace in the 1970s. Most of the larger computer science departments were linked by the ARPANET (an experimental defense department network linking research projects funded by ARPA, the defense advanced research projects agency). By the mid 1970s, most university computer centers were linked by BITNET, and as Unix emerged from Bell Labs, most Unix sites joined UUNET, an informal network of Unix sites, originally linked by dial-up lines. E-mail became the "killer app" on all of these early networks.

The 1980's

Personal computers such as the Apple II and the first IBM PCs came with systems that were typical of the early disk operating systems. IBM even called its system PCDOS, and many PC users just called it DOS, as if no other system had ever had this name. Eventually, it emerged that this was a Microsoft product, although IBM originally sold it without this identification.

In the 1980's, Unix was pried free from Bell Labs and AT&T. The University of California developed BSD Unix, originally under AT&T license, but they reimplemented enough of it that, as time passed, BSD Unix was wrested free of AT&T. Linus Torvalds, a Finnish hacker, developed another Unix clone, Linux, and many manufacturers, under AT&T license, commercialized their own UNIX variants. IBM developed AIX. HP developed HPUX. Sun developed Solaris.

Unix was adapted to run on multiprocessors by two competing vendors in this decade, Sequent and Encore. In general, Unix worked very well on machines with on the order of 16 CPUs. Earlier operating systems from Burroughs Corporation, the University of Michigan and Carnegie Mellon Univeristy had demonstrated similar performance in earlier decades.

Independently of all this, Carnegie Mellon university developed a system called Mach that was supposed to be used as a replacement kernel under Unix, but was in fact far more. Mach would be seen as an academic curiosity for many years, but eventually, BSD was rebuilt on top of a Mach kernel, and eventually, Apple would chose BSD/Mach as the foundation for MacOS X.

Window managers, first developed in the 1970's at SRI and Xerox PARC, came into maturity in the 1980's. Window managers don't need to rest on sophisticated operating system technology. Prior to Windows NT and 95 from Microsoft and MacOS X from Apple, the dominant commercial window managers sat on top of rather primitive disk operating systems. In contrast, however, the X window system from MIT, developed under BSD Unix, took complete advantage of the available operating system technology. X remains the dominant window manager on Unix and related platforms.

In the early 1980's, mail gateways were installed between the existing computer networks, creating serious headaches that were resolved later in the decade by the creation of the Internet, a generalization of the old ARPANET. By this time, operating systems such as Unix provided a good suite of network access primitives, and for most of the decade, the most common Internet servers were Digital Equipment Corporation's VAX computers running BSD Unix.

The 1990's

In the 1990's, the personal computing field finally rose above the level of DOS. Windows NT emerged when Microsoft bought the ashes of DEC's VAX VMS operating system development group. Windows 95 was a reaction to this, but both are real operating systems in the sense that emerged in the 1960's. MacOS X is another solid operating sysem to reach the desktop in this era. These systems were finally mature enough to incorporate decent support for virtual memory and network connectivity.

By the end of the 1990s, the operating systems running on typical desktop computers were as complex as any mainframe operating system of the 1960's. Essentially all of the innovative ideas from systems such as Multics were to be found on desktop and laptop computers, complete with full support for networking and window management.

The 2000's

As with the minicomputer revolution and the microcomputer revolutions before, the mobile communications revolution created a wide open niche in which, at least initially, competition flourished. Initially, each cellphone and PDA vendor based their product on proprietary systems, but as the market grew and the function of PDAs and Cellphones began to merge, two systems emerged as the primary competitors in this new niche, Windows CE and Android.

Neither of these represents any kind of revolutionary new approach to operating systems. Windows CE is a direct descendant of Windows, freed from dependency on the Intel x86 family and stripped of the baggage of "integration" with a full suite of office productivity tools. Android, in turn, is based on Linux, stripped of the assumption that the shell or a window manager will be the primary application launchers.

A third thread has woven into both of these systems, and that is the desire of many major players to lock down the system, controlling what applications the user is permitted to launch, what files the user is permitted to store, and where control is not possible, allowing for pervasive monitoring of the actions taken by users of mobile platforms. This has led to innovations such as trusted platform monitors, but it also raises serious questions about the relationship between operating system developers and civil liberties. Is it ethical for programmers to write code that permits pervasive surveillance of cellphone users? Probably not.

History of networking

The precursor to the Internet was jumpstarted in the early days of computing history, in 1969 with the U.S. Defense Department's Advanced Research Projects Agency Network (ARPANET). ARPA-funded researchers developed many of the protocols used for Internet communication today. This timeline offers a brief history of the Internet's evolution:

1934: Belgian information expert named Paul Otlet imagined a "Radiated Library" that would use technology of the day — the telephone and radio — to create something very much like the Internet.

1965: Two computers at MIT Lincoln Lab communicate with one another using packet-switching technology.

1968: Beranek and Newman, Inc. (BBN) unveils the final version of the Interface Message Processor (IMP) specifications. BBN wins ARPANET contract.

A visualization of Internet connections in the United States. The lines represent connections between routers in major urban areas throughout the country.

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1969: On Oct. 29, UCLA's Network Measurement Center, Stanford Research Institute (SRI), University of California-Santa Barbara and University of Utah install nodes. The first message is

“LO,” which was an attempt by student Charles Kline to “LOGIN” to the SRI computer from the university. However, the message was unable to be completed because the SRI system crashed.

1972: BBN’s Ray Tomlinson introduces network email. The Internetworking Working Group (INWG) forms to address need for establishing standard protocols.

1973: Global networking becomes a reality as the University College of London (England) and Royal Radar Establishment (Norway) connect to ARPANET. The term Internet is born.

1974: The first Internet Service Provider (ISP) is born with the introduction of a commercial version of ARPANET, known as Telenet.

1974: Vinton Cerf and Bob Kahn (the duo said by many to be the Fathers of the Internet) publish "A Protocol for Packet Network Interconnection," which details the design of TCP.

1976: Queen Elizabeth II hits the “send button” on her first email.

1979: USENET forms to host news and discussion groups.

1981: The National Science Foundation (NSF) provided a grant to establish the Computer Science Network (CSNET) to provide networking services to university computer scientists.

1982: Transmission Control Protocol (TCP) and Internet Protocol (IP), as the protocol suite, commonly known as TCP/IP, emerge as the protocol for ARPANET. This results in the fledgling definition of the Internet as connected TCP/IP internets. TCP/IP remains the standard protocol for the Internet.

1983: The Domain Name System (DNS) establishes the familiar .edu, .gov, .com, .mil, .org, .net, and .int system for naming websites. This is easier to remember than the previous designation for websites, such as 123.456.789.10.

1984: William Gibson, author of "Neuromancer," is the first to use the term "cyberspace."

1985: Symbolics.com, the website for Symbolics Computer Corp. in Massachusetts, becomes the first registered domain.

1986: The National Science Foundation’s NSFNET goes online to connected supercomputer centers at 56,000 bits per second — the speed of a typical dial-up computer modem. Over time the network speeds up and regional research and education networks, supported in part by NSF, are connected to the NSFNET backbone — effectively expanding the Internet throughout the United States. The NSFNET was essentially a network of networks that connected academic users along with the ARPANET.

1987: The number of hosts on the Internet exceeds 20,000. Cisco ships its first router.

1989: World.std.com becomes the first commercial provider of dial-up access to the Internet.

1990: Tim Berners-Lee, a scientist at CERN, the European Organization for Nuclear Research, develops HyperText Markup Language (HTML). This technology continues to have a large impact on how we navigate and view the Internet today.

1991: CERN introduces the World Wide Web to the public.

1992: The first audio and video are distributed over the Internet. The phrase “surfing the Internet” is popularized.

1993: The number of websites reaches 600 and the White House and United Nations go online. Marc Andreessen develops the Mosaic Web browser at the University of Illinois, Champaign-Urbana. The number of computers connected to NSFNET grows from 2,000 in 1985 to more than 2 million in 1993. The National Science Foundation leads an effort to outline a new Internet architecture that would support the burgeoning commercial use of the network.

1994: Netscape Communications is born. Microsoft creates a Web browser for Windows 95.

1995: Compuserve, America Online and Prodigy begin to provide Internet access. Amazon.com, Craigslist and eBay go live. The original NSFNET backbone is decommissioned as the Internet’s transformation to a commercial enterprise is largely completed.

1996: The browser war, primarily between the two major players Microsoft and Netscape, heats up. CNET buys tv.com for \$15,000.

1997: PC makers can remove or hide Microsoft’s Internet software on new versions of Windows 95, thanks to a settlement with the Justice Department. Netscape announces that its browser will be free.

1998: The Google search engine is born, changing the way users engage with the Internet.

1999: AOL buys Netscape. Peer-to-peer file sharing becomes a reality as Napster arrives on the Internet, much to the displeasure of the music industry.

2000: The dot-com bubble bursts. Web sites such as Yahoo! and eBay are hit by a large-scale denial of service attack, highlighting the vulnerability of the Internet. AOL merges with Time Warner.

A newly expanded global Internet, to focus solely on science and education, now includes half of the world’s countries. The high-speed fiber-optic network connects users at speeds of 10 Gbps.

A newly expanded global Internet, to focus solely on science and education, now includes half of the world’s countries. The high-speed fiber-optic network connects users at speeds of 10 Gbps.

2001: A federal judge shuts down Napster, ruling that it must find a way to stop users from sharing copyrighted material before it can go back online.

2003. The SQL Slammer worm spread worldwide in just 10 minutes. Myspace, Skype and the Safari Web browser debut.

2004: Facebook goes online and the era of social networking begins. Mozilla unveils the Mozilla Firefox browser.

2005: YouTube.com launches.

2006: AOL changes its business model, offering most services for free and relying on advertising to generate revenue. The Internet Governance Forum meets for the first time.

2009: The Internet marks its 40th anniversary.

2010: Facebook reaches 400 million active users.

Pioneers of computing

Person	Contribution
<u>al-Khwārizmī</u>	Coined word Algorithm
<u>John Atanasoff</u>	Built the first electronic digital computer, the <u>Atanasoff–Berry Computer</u> , though it was neither programmable nor <u>Turing-complete</u> .
<u>Charles Babbage</u>	Originated the concept of a programmable general-purpose computer. Designed the <u>Analytical Engine</u> and built a prototype for a less powerful <u>mechanical calculator</u> .
<u>John Backus</u>	Invented <u>FORTRAN</u> (<i>Formula Translation</i>), the first practical high-level programming language, and he formulated the <u>Backus–Naur form</u> that described the formal language <u>syntax</u> .
<u>Jean Bartik</u>	One of the first <u>computer programmers</u> , on <u>ENIAC</u> (1946), an early <u>Vacuum tube computer</u> back when "programming" involved using cables, dials, and switches to physically rewire the machine. Worked with <u>John Mauchly</u> toward <u>BINAC</u> (1949), <u>EDVAC</u> (1949), <u>UNIVAC</u> (1951) to develop early " <u>Stored program</u> " computers.
<u>Tim Berners-Lee</u>	Invented <u>worldwide web</u> . With <u>Robert Cailliau</u> , sent first HTTP communication between client and server.
<u>George Boole</u>	Formalized <u>Boolean algebra</u> , the basis for <u>digital logic</u> and computer science.
<u>Per Brinch Hansen</u>	Developed the <u>RC 4000 multiprogramming system</u> which introduced the concept of an <u>operating system kernel</u> and the separation of policy and mechanism, effectively the first <u>microkernel</u> architecture. ^[4] Co-developed the <u>monitor</u> with <u>Tony Hoare</u> , and created the first monitor implementation. ^[5] Implemented the first form of <u>remote procedure call</u> in the RC 4000, ^[4] and was first to propose remote procedure calls as a structuring concept for <u>distributed computing</u> . ^[6]
<u>Vint Cerf</u>	With <u>Bob Kahn</u> , designed the <u>Transmission Control Protocol</u> and <u>Internet Protocol</u> (TCP/IP), the primary data <u>communication protocols</u> of the <u>Internet</u> and other <u>computer networks</u> .
<u>Noam Chomsky</u>	Made contributions to computer science with his work in <u>Linguistics</u> . He developed <u>Chomsky hierarchy</u> , a discovery which has directly impacted <u>Programming language theory</u> and other branches of computer science.

Person	Contribution
<u>Edgar F. Codd</u>	Proposed and formalized the <u>relational model</u> of data management, the theoretical basis of <u>relational databases</u> .
<u>Ole-Johan Dahl</u>	With <u>Kristen Nygaard</u> , invented the proto-object oriented language <u>SIMULA</u> .
<u>J. Presper Eckert</u>	With <u>John Mauchly</u> , designed and built the <u>ENIAC</u> , the first modern (all electronic, Turing-complete) computer, and the <u>UNIVAC I</u> , the first commercially available computer.
<u>Douglas Engelbart</u>	Best known for inventing the computer mouse (in a joint effort with Bill English); as a pioneer of human-computer interaction whose Augment team developed <u>hypertext</u> , <u>networked computers</u> , and precursors to <u>GUIs</u> .
<u>Tommy Flowers</u>	Designed and built the Mark 1 and the ten improved Mark 2 <u>Colossus computers</u> , the world's first programmable, digital, electronic, computing devices.
<u>Gottlob Frege</u>	Developed <u>first-order predicate calculus</u> , which was a crucial precursor requirement to developing computation theory.
<u>Grace Hopper</u>	Pioneered work on the necessity for high-level programming languages, which she termed <i>automatic programming</i> , and wrote the <u>A-O compiler</u> , which heavily influenced the <u>COBOL</u> language.
<u>Cuthbert Hurd</u>	Helped the International Business Machines Corporation develop its first general-purpose computer, the <u>IBM 701</u> .
<u>Kenneth E. Iverson</u>	Assisted in establishing the first graduate course in computer science (at <u>Harvard</u>) and taught that course; invented the <u>APL programming language</u> and made contribution to interactive computing.
<u>Joseph Marie Jacquard</u>	Built and demonstrated the <u>Jacquard loom</u> , a programmable mechanized loom controlled by punch cards.
<u>Maurice Karnaugh</u>	Inventor of the <u>Karnaugh map</u> , used for logic function minimization.
<u>Jacek Karpinski</u>	Developed the first differential analyzer that used transistors, and developed one of the first machine learning algorithms for character and image recognition. Also was the inventor of one of the first minicomputers, the <u>K-202</u> .
<u>Ada Lovelace</u>	Began the study of scientific computation, analyzing Babbage's work in her <i>Sketch of the Analytical Engine</i> , and was the namesake for the <u>Ada programming language</u> .
<u>John Mauchly</u>	With <u>J. Presper Eckert</u> , designed and built the <u>ENIAC</u> , the first modern (all electronic, Turing-complete) computer, and the <u>UNIVAC I</u> , the first commercially available computer. Also worked on <u>BINAC</u> (1949), <u>EDVAC</u> (1949), <u>UNIVAC</u> (1951) with Grace Hopper and Jean Bartik, to develop early " <u>Stored program</u> " computers.
<u>John McCarthy</u>	Invented <u>LISP</u> , a <u>functional programming language</u> .

Person	Contribution
<u>Marvin Minsky</u>	Co-founder of <u>Artificial Intelligence Lab</u> at <u>Massachusetts Institute of Technology</u> , author of several texts on AI and philosophy.
<u>John von Neumann</u>	Formulated the <u>von Neumann architecture</u> upon which most modern computers are based.
<u>Blaise Pascal</u>	Invented the <u>mechanical calculator</u> .
<u>Dennis Ritchie</u>	With <u>Ken Thompson</u> , pioneered the <u>C programming language</u> and the <u>Unix</u> computer operating system at Bell Labs.
<u>Claude Shannon</u>	Founded <u>information theory</u> and practical digital circuit design.
<u>Alan Turing</u>	Made several founding contributions to computer science, including the <u>Turing machine</u> computational model, the conceiving of the stored program concept and the designing of the high-speed <u>ACE</u> design. Widely considered as the father of <u>Computer Science</u> and <u>Artificial Intelligence</u> .
<u>Konrad Zuse</u>	Built the first digital freely programmable computer, the <u>Z1</u> . Built the first functional tape-stored program-controlled computer, the <u>Z3</u> . The Z3 was proven to be Turing-complete in 1998. Produced the world's first commercial computer, the <u>Z4</u> . Designed the first high-level programming language, <u>Plankalkül</u> .
Larry Page and Sergey Brin	Developed Google search engine
Mark Zuckerberg	Developed Social media facebook
Larry Ellison	Developed could computing